BUILDING CONSTRUCTION

1. PREPARATION FOR CONSTRUCTION

1.1. STAKING-OUT THE BUILDING (Mercado)

Methods in staking-out the building:

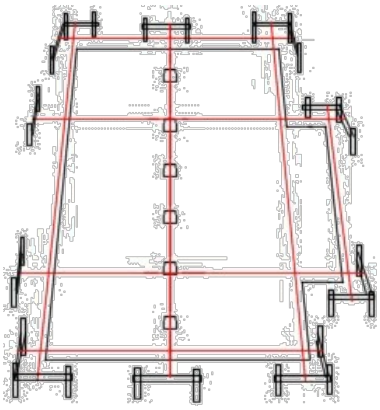
1. Measure the required setback from the front corner monuments and drive two stakes and stretch a line between them to represent the front building line.

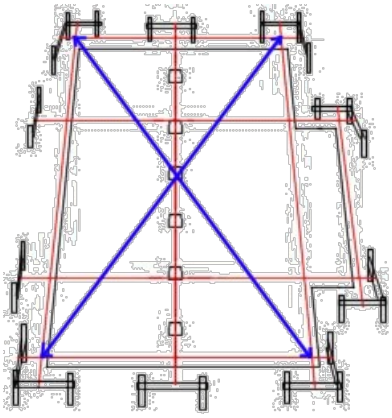
2. Measure the required side setback from one of the side lot lines along the line established in (1) and drive a stake A which will represent the first corner of the building. Then from Stake A, measure the width of the building and mark with a stake B to obtain the other front corner.

3. Estimate right angles from stakes A and B and measure the length of the building. In those two points, drive two temporary stakes C and D which will mark the rear corners of the building. C-D should be equal to A-B.

4. Draw lines along the diagonals A-D and B-C and ensure that these are equal.

5. Transfer the building lines to batter boards.





1.2. LAYING THE BATTERBOARDS (Mercado)

After staking out, transfer the building lines to batter boards. Batter boards are horizontal boards used to establish the height of the finish foundation and to support the guidelines for the excavation of footing trenches. They should be carefully leveled with a carpenter's level or a transit and the height must be a little higher that the top of the finished foundation.





1.3. FORMWORK AND SHORING

Fresh concrete must be shaped and supported by **formwork** until it cures and can support itself. To support beam and slab forms while concrete is being poured and cured, temporary supports called **shoring** are used. Materials for forms and shoring construction are (1) Lumber and (2) Steel.

1. Lumber Forms and Shoring

Lumber used in form construction should only be partially seasoned and to some extent slightly wetted in order to prevent swelling and distortion of the forms.

Sheet lumber are dressed at least one side and both edges even for non-exposed surfaces, as the removal and cleaning of the forms are greatly facilitated thereby. In face work, where smooth and true surface is quite important, the lumber should be dressed on all four sides.

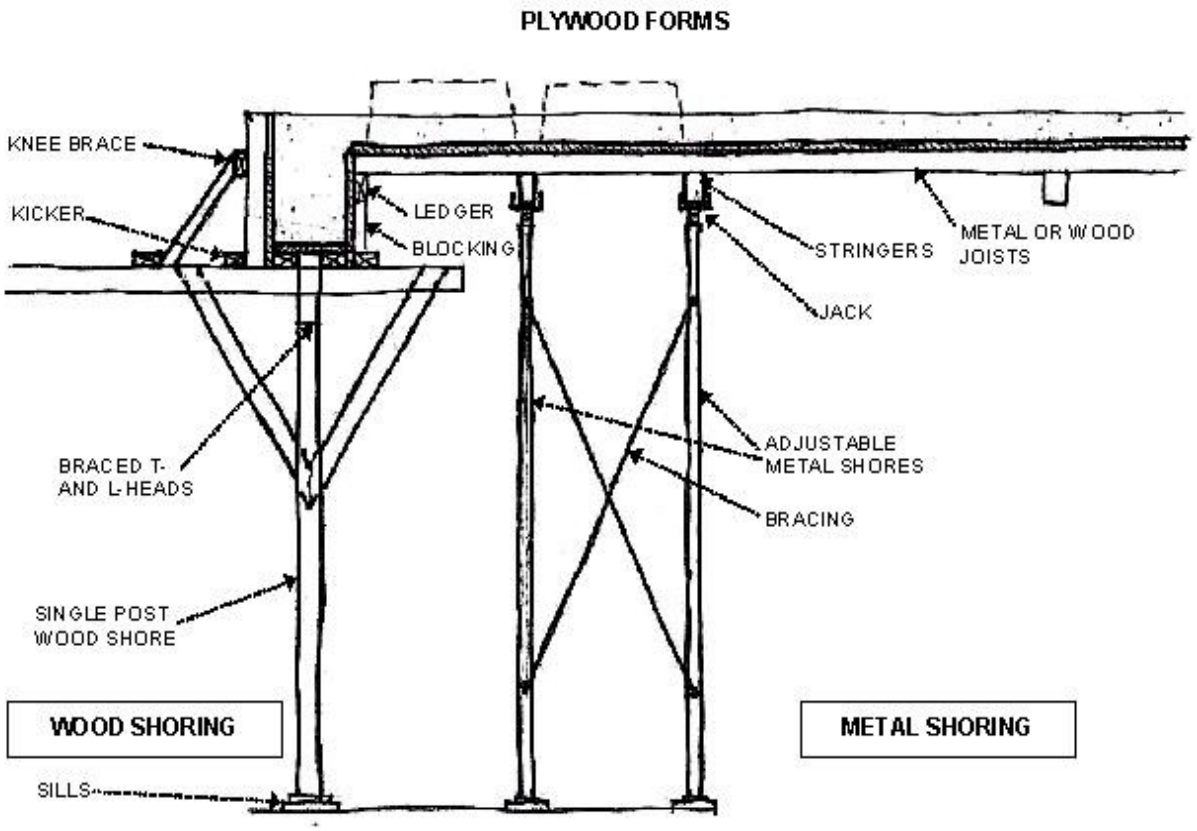
In floor and wall panels, tight joints between boards may be obtained by using tongue-and grooved stock. Joints in forms for columns, beams, and girders are sometimes made tight by simply dressing the lumber true to edge, forming square of butt joints.

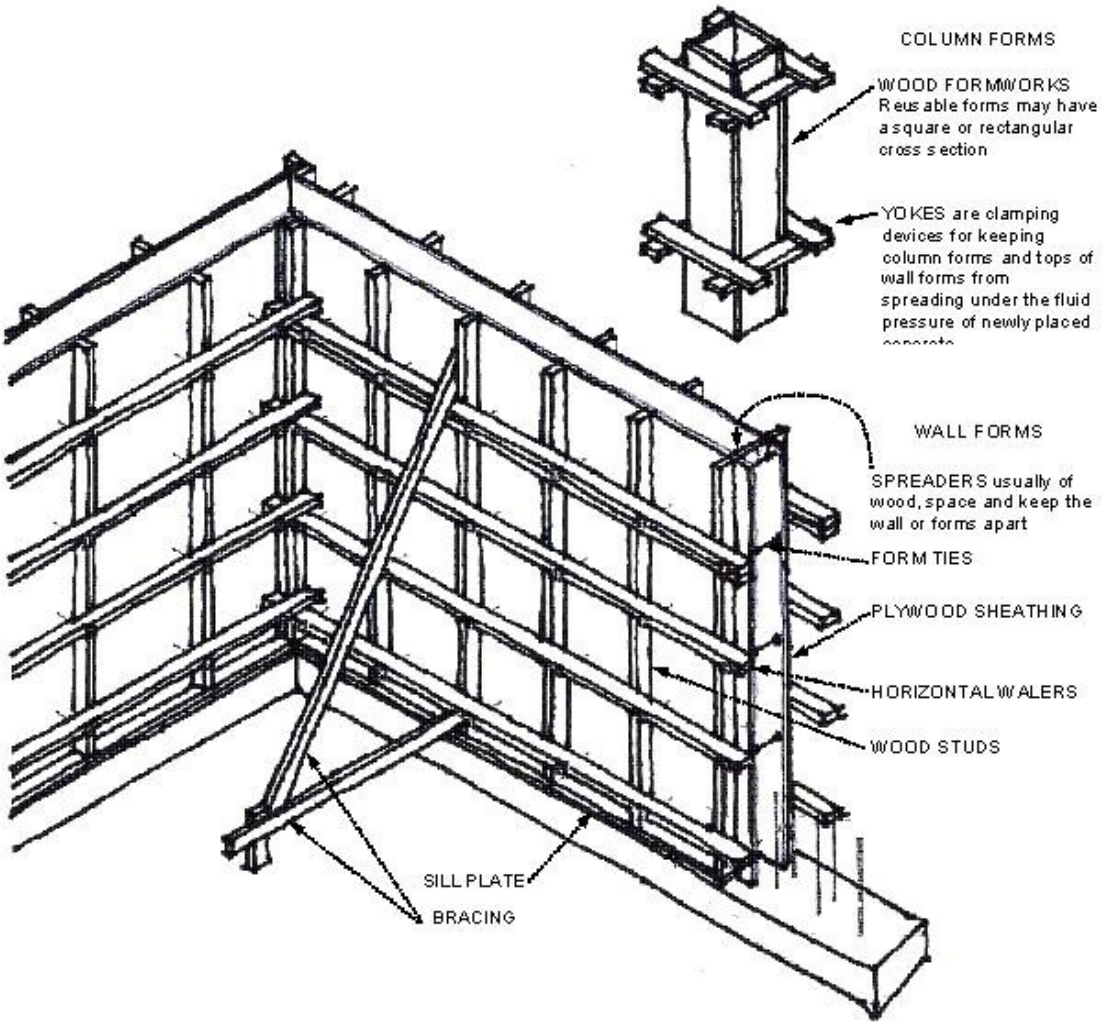
Sizes of lumber frequently used are: 2-in.stock for columns, beams and girder bottoms; 1-in. stock for floor panels and beam and girder sides; 2x4s for struts, posts, shores, and uprights; 1 or 2-inch stock for cleats.

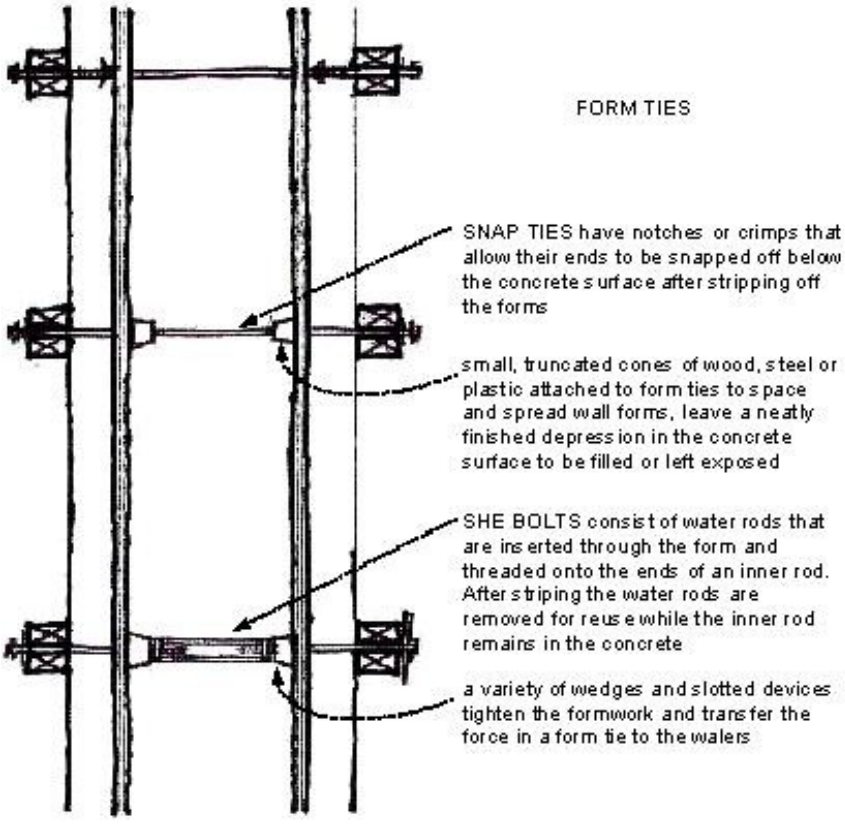
Crude oil and petroline are used to prevent concrete from adhering to the wood and preserve the forms against damage by alternate wetting and drying. The forms should preferably be oiled before they are set in place. Oil should not be used, however, on forms against surfaces which are to be plastered, as oil prevents adhesion of the plaster. Wetting with water in such cases will be sufficient.

To hold wall forms together, wire ties or bolts and rods may be used. The latter are preferred and should be arranged that upon removal of the forms, no metal shall be within one inch of any surface. Wire ties should be used only on light and unimportant work where discoloring will not be objectionable.

Plywood Forms. These are used where a smooth surface is required. The plywood should be waterproof, Grade “A” and at least ½” thick.




COLUMN FORMS
WOOD FORMWORKS
Reusable forms may have a square or rectangular cross section
YOKES are clamping devices for keeping column forms and tops of wall forms from spreading under the fluid pressure of newly placed
WALL FORMS
SPREADERS usually of wood, space and keep the wall or forms apart
FORM TIES
PLYWOOD SHEATHING
HORIZONTAL WALERS
WOOD STUDS
SILL PLATE
BRACING


FORM TIES
SNAP TIES have notches or crimps that allow their ends to be snapped off below the concrete surface after stripping off the forms
small, truncated cones of wood, steel or plastic attached to form ties to space and spread wall forms, leave a neatly finished depression in the concrete surface to be filled or left exposed
SHE BOLTS consist of water rods that are inserted through the form and threaded onto the ends of an inner rod. After striping the water rods are removed for reuse while the inner rod remains in the concrete
a variety of wedges and slotted devices tighten the formwork and transfer the force in a form tie to the walers

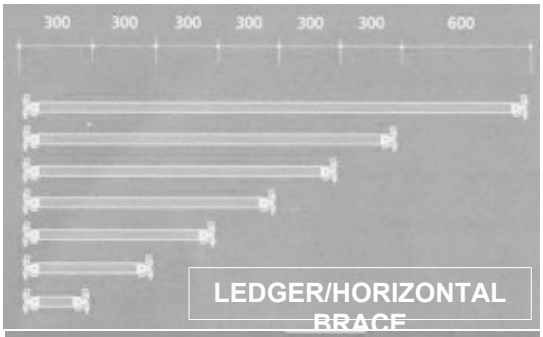
2. Steel Forms and Shoring



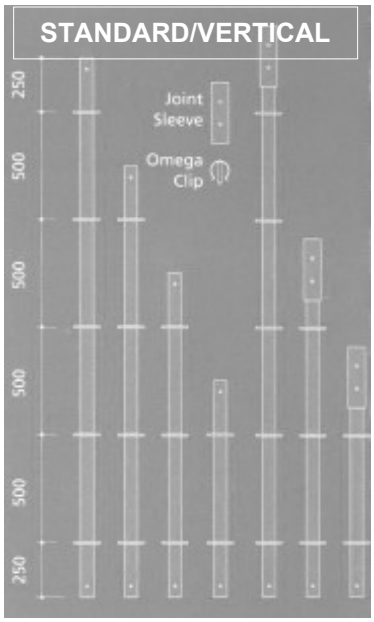
Steel forms may be in the form of pans for concrete joist construction or steel decking or corrugated steel for concrete slabs and slab-and-joist construction.

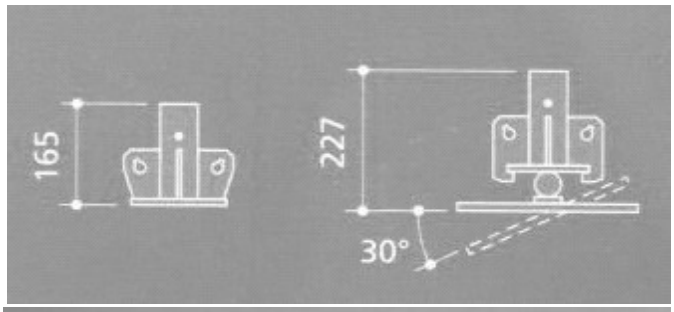
The major components of metal shoring or scaffolding are:

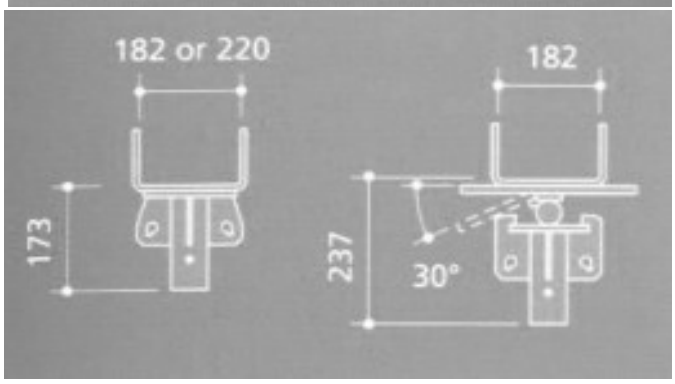
a) The ledger or the horizontal brace.
b) The brace or the diagonal. This component may be of the adjustable and the fixed type.
c) The standard or the vertical component.
d) Accessories: heads, jacks and bases.

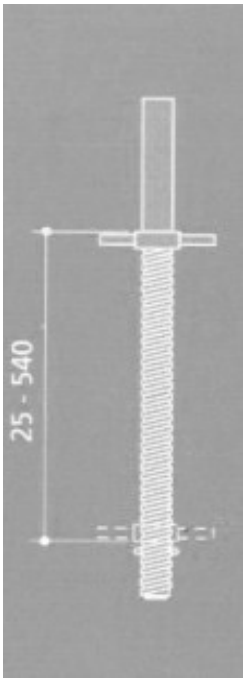












1.4. CONSTRUCTION TOOLS & EQUIPMENT

Tools and equipment employed in construction are grouped into four major categories:

1. **HAND TOOLS** are the tools that use power delivered by man only.

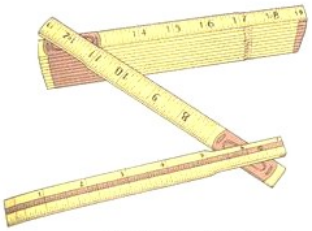
   A pry bar is used to force open boards used in forming concrete.



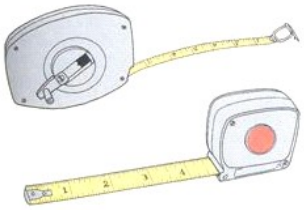
Pry bar.

Measurement and layout tools are the following:

(a) Folding rule and tape measure are the most common tools for measuring boards, pipe, wire



Folding rule.

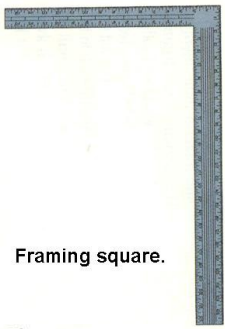


Tape measures.

(b) Digital rule is used to measure relatively long distances such as those in highway construction.



Digital rule.



Framing square.

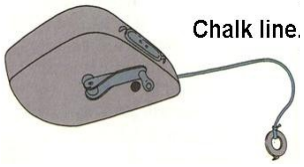
(c) Framing square is a layout tool that is used to measure 90-degree angles at the corners of framework and joints. They can also be employed to determine cutting angles on dimension lumber.

(d) Level is a long, straight tool that contains one or more vials of liquid and used to determine if the horizontal or vertical is exact.



Level.

(e) Chalk line or chalk box is used for marking lines.



Chalk line.

Types of **hammers** are the following:

(a) Claw hammer is an ordinary hammer used to drive or remove nails.



Claw hammer.

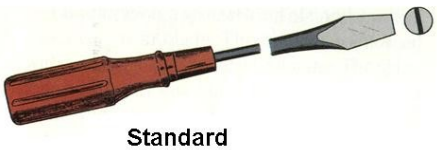
(b) Sledgehammer is a heavy hammer used to drive stakes into the ground and to break up concrete and stone.



Sledgehammer.

Types of **screwdrivers** are:

(a) Standard screwdriver has a flat tip and is designed to fit a standard slotted screw.



Standard screwdriver.

(b) Phillips screwdriver has an X-shaped tip and is used to turn Phillips-head screws only.



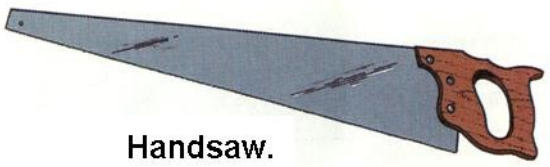
Phillips screwdriver.

(c) Spiral ratchet screwdriver is that which relies on a pushing force rater than a twisting force.



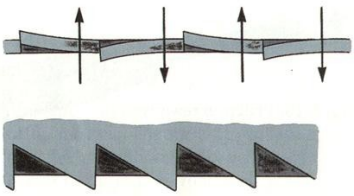
Spiral ratchet screwdriver

Types of **saws** are the following:



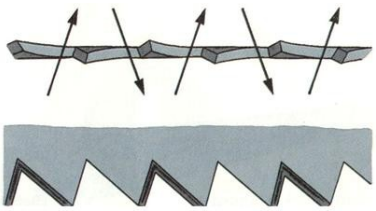
Handsaw.

(a) Ripsaw has chisel-like teeth designed for ripping or cutting with the grain of wood.



Ripsaw blade. Top, top view. Bottom, side view.

(b) Crosscut saw is used to cut across the grain of wood.



Crosscut-saw blade. Top, top view. Bottom, side view.

(c) Backsaw is a special type of handsaw that has a very thin blade and makes very straight cuts such as those on trims and mouldings.

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(d) Hacksaw is used to cut metals.

Types of **Chisels**:

(a) Wood chisel is used to trim wood and clear away excess material from wood joints.


Wood chisel.

(b) Cold chisel is used to trim metals.

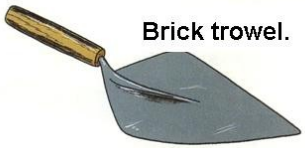

Cold chisel.

Types of specialized hand tools are the following:


Nail set.


Bull float.


Stillson (pipe) wrench.

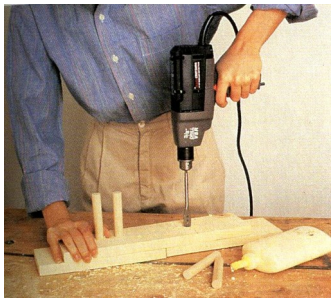

Brick trowel.


Blind riveter.

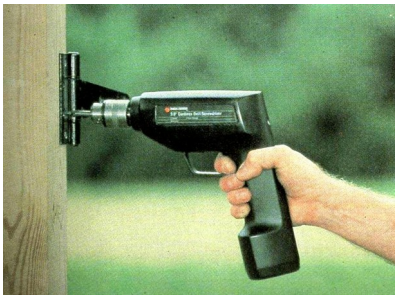
(a) Nail set is used to drive finishing nails below the surface of a wooden trim or molding.
(b) Pipe wrench is used to turn round objects like pipes.
(c) Brick trowel is used to place and trim mortar between bricks or concrete blocks.
(d) Bull float is used to smoothen out the surface of wet concrete.

(e) Blind riveter is used to fasten pieces of sheet metal together.

2. **POWER TOOLS** are those that employ power supplied by forces other than that coming from humans.



POWER DRILL

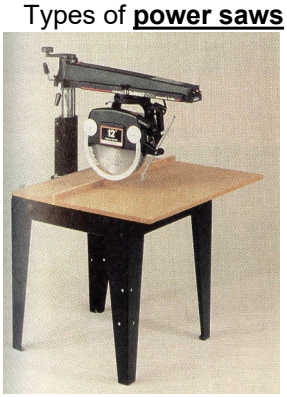


POWER SCREWDRIVER/SCREWGUN

Power drill is used to drill holes in wood, metal and concrete.

Power screwdriver or screwgun is used to install and remove screws.

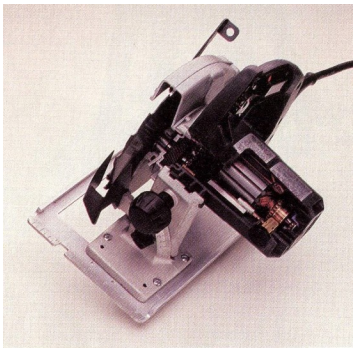
Types of **power saws** are the following:



RADIAL ARM



TABLE



PORTABLE CIRCULAR



POWER MITER



SABER

(a) Radial arm saw is used for crosscutting wood and consists of a motor-driven saw blade that is hung on an arm over a table.

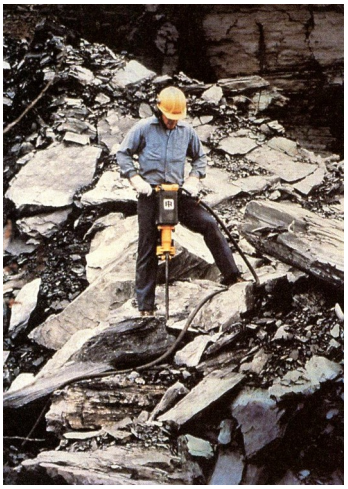
(b) Table saw is used for cutting large sheets of wood and wood composites and consists of a blade mounted on an electric motor beneath a table-like surface.

(c) Portable circular saw is used for cutting materials that are difficult to cut with stationary tools.

(d) Power miter saw is a circular saw mounted over a small table used to cut various angles in wood.

(e) Saber saw is used to cut curves or holes in floors and roofs for pipes and has a small knife-shaped blade that moves up and down.

Type of **power hammers** are:



PNEUMATIC/JACKHAMMER



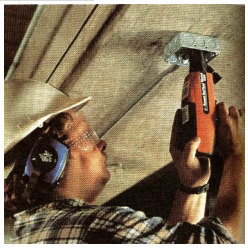
ROTARY HAMMER

(a) Pneumatic hammer or jackhammer is used to break up concrete or asphalt paving.
(b) Rotary hammer is like an electric drill that operates with both rotating and reciprocating actions and is used to drill holes in concrete.

Type of **power nailers and staplers** are the following:



NAILERS/NAIL GUN



STUD DRIVER

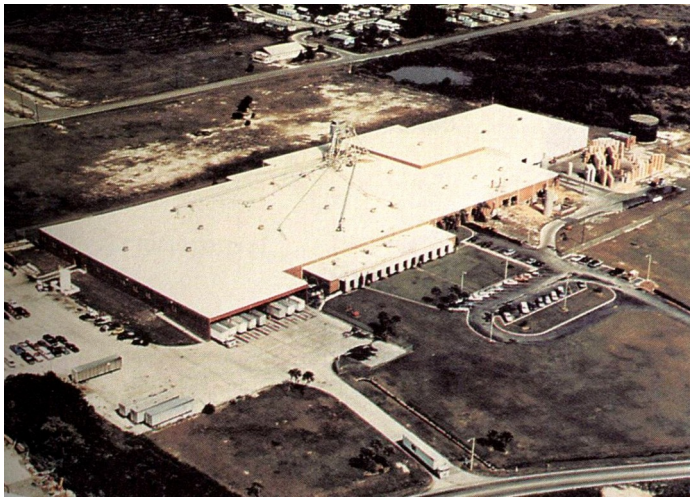


STAPLER

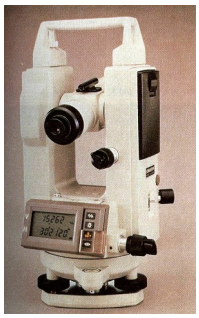
(a) Nailers or nail guns fasten materials together by shooting nails into the building material.
(b) Powder-actuated stud driver is a kind of nailer that is powered by gunpowder and is used to drive long pins into wood, steel or concrete.
(c) Staplers are like nailers but are loaded with u-shaped staples instead of nails for fastening.

3. **CONSTRUCTION EQUIPMENT** is a term that refers to large, complex tools and machines that is designed to do a particular job.

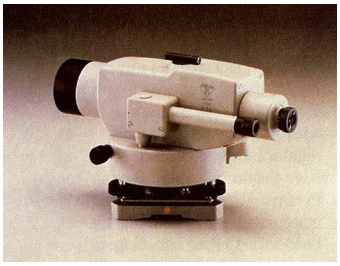
Conveyor is an equipment which moves materials other than fluids.



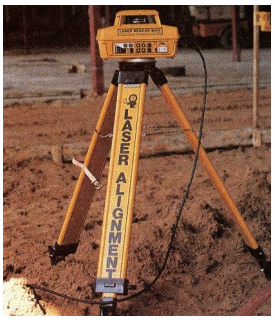
Types of **surveying equipment**:



TRANSIT



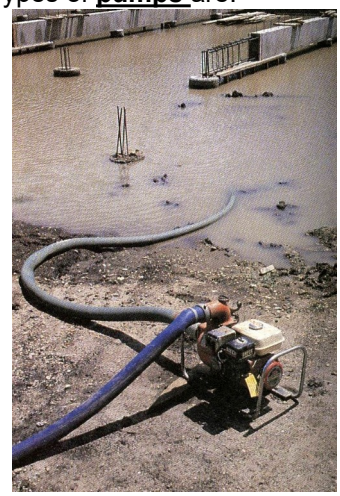
SURVEYOR'S LEVEL



CONSTRUCTION LASER

(a) Transit is an equipment used by surveyors to measure horizontal and vertical angles to obtain land elevation.
(b) Surveyor's level is that which to determine an unidentified elevation from a known one.
(c) Construction laser is flashes a narrow, accurate beam of light to make a baseline for additional measurements and is used as a level or as an alignment tool.

Types of **pumps** are:



WATER PUMP

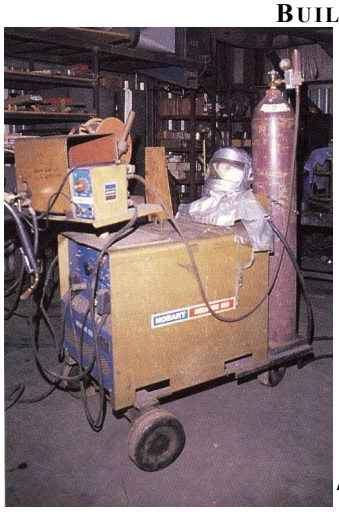


CONCRETE PUMP

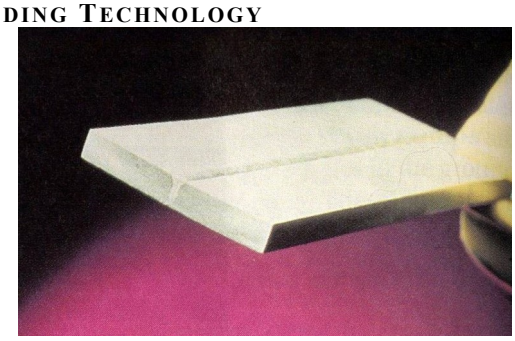
(a) Water pump is used to pump water out of holes in the ground so that construction work can commence.

(b) Concrete pump is used to move concrete from the concrete mixer to the concrete form.

Types of **welding machines** are the following:



ARC WELDING MACHINE



WELD MADE BY A LASER
POWERED LASER

(a) Arc welding machine is used to weld materials by melting portions of the metal.
(b) Laser-powered welder is used to weld material by employing a laser to heat the metal.

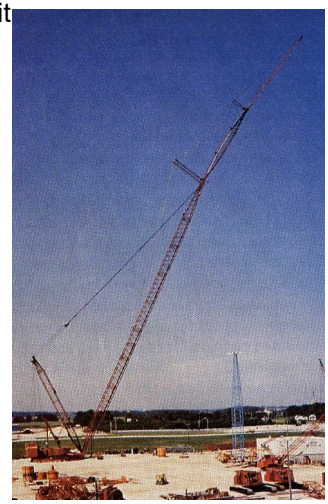
4. **HEAVY EQUIPMENT** is equipment which is very large and very powerful.
Bulldozer is a tractor with a pushing blade which moves earth and clears land of bushes and trees.



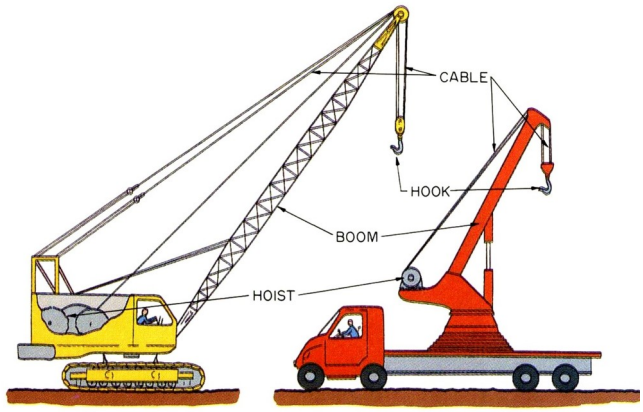
Cranes are machines that lift large and heavy materials. The types of cranes are:

640-foot crane

(a) Crawler crane is a crane mounted on metal treads so that it can move over rough terrain.



(b)



MECHANICAL CRAWLER CRANE HYDRAULIC TRUCK CRANE

Truck crane is mounted on a truck frame so that it can be driven in the site.
Tower crane or climbing crane is used in the construction of tall building because it has a built-in jack that raises the crane from floor to floor as the building is constructed.

Excavator is a machine used for digging or scooping earth from a place and depositing it in another. Types of excavators are:



(a) Backhoe is used for general digging which is usually mounted on either a crawler or truck frame.



(b) Trencher is a special kind which digs trenches or long, narrow ditches for pipelines or cables.

(c) Front-end loader is a large shoveling machine that can scoop or deposit a large amount of material.



Highway construction equipment are:

Scraper is a machine that loads, hauls and dumps soil over medium to long distances.



Grader is an earthworking machine that grades or levels the ground.



Compactor or roller is a machine that compacts soil to prepare for road paving.

Paver is a machine that places, spreads and finishes concrete or asphalt paving material.

1.5. FOUNDATION BED

The word “foundation” is applied to:

1. Construction below grade, such as footing courses, basement walls, etc., forming the lower section of a structure;
2. The natural material, the particular part of the earth’s surface on which the construction rests;
3. Special construction such as piling or piers used to transmit the loads of the building to firm substrata.

Foundation bed is the natural material on which the construction rests. Walls, piers and columns below grade are called, in general, foundation walls, piers and columns to distinguish them from similar construction above grade. The lower portions of walls, piers or columns which are spread to provide a safe base are called footing courses.

1.5.1. TYPES OF FOUNDATION BEDS

Foundation beds may be classified as follows:

1. Rock (solid rock, bedrock or ledge). Undisturbed rock masses forming an undisturbed part of the original rock-formation. Some examples of harder rocks are granite, slate, sandstone, and limestone which are all capable of carrying the load of any ordinary structure. Examples of softer rocks are shale, shaley slates and certain marley limestone and clay stones.
2. Decayed rock (rotten rock). Sand, clays and other materials resulting from the disintegration of rock masses, lacking the coherent qualities but occupying the space formerly occupied by the original rock.
3. Loose rock. Rock masses detached from the ledge of which they originally formed a part.
4. Gravel. Detached rock particles, generally water-worn, rounded and intermediate in size between sand particles and boulders.
5. Boulders. Detached rock masses larger than gravel, generally rounded and worn as a result of having been transported by water a considerable distance from the ledges of which they originally formed a part.
6. Sand. Non-coherent rock particles smaller than ¼" in maximum dimension.
7. Clay. A plastic material resulting from the decomposition and hydration of feldspathic rocks, being hydrated silicate of alumina, generally mixed with powdered feldspar, quartz and other materials.
8. Hard-pan. Any strong coherent mixture of clay or other cementing material with sand, gravel and boulders.
9. Silt. A finely divided earthy material deposited from running water.
10. Mud. Finely divided earthy material generally containing vegetable matter and deposited from still or slowly moving water.
11. Mould. Earthy material containing a large proportion of humus or vegetable matter.
12. Loam. Earthy material containing a proportion of vegetable matter.
13. Peat. Compressed and partially carbonized vegetable matter.
14. Filled Ground. All artificial fills and some natural fills are liable to a more or less uniform but continuous settlement or shrinkage due to the gradual consolidation of the material of which the fill is composed.

1.5.2. ALLOWABLE LOADS ON FOUNDATION BEDS

Because there are variations in the materials and conditions affecting such materials, thorough investigation is required before one can determine the allowable unit load on the foundation bed. In cases wherein the material and conditions are uniform over the entire site of the building, a uniform unit load may be used. In cases wherein entirely different conditions exist under different portions of the same building, the unit load on the foundation bed must be reduced as much as possible so as to reduce the differences in settlements between the two sections of the building to a minimum.

1.6. SITE INVESTIGATION

Before any design is made, the architect is required to get as much valuable data about site excavation and building erection at the project location in order to be able to determine the character of the materials which will be encountered at the level of a foundation bed.

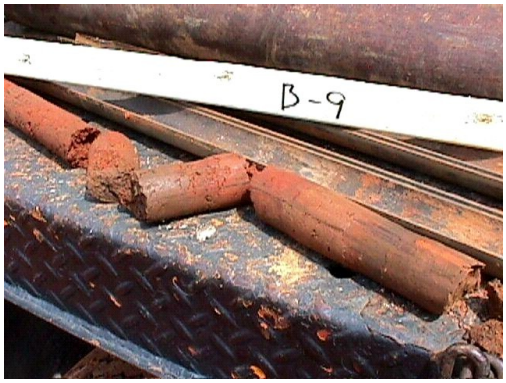
1.6.1. METHODS OF EXPLORATION

1. Test pits. For shallow work, an open pit is the most suitable method since it calls for an actual inspection of the undisturbed material over a considerable area.

2. Test borings. For excavations that are carried no deeper than the proposed level, the underlying material may be investigated by test boring.



SOIL BORING RIG



SPT SAMPLE

1.6.2. LOADING TESTS

Loading tests of the materials forming the foundation bed are made to assist in determining its safe bearing capacity.

1.7. EXCAVATION AND EARTHWORKING

These processes entail the following:



1. Excavating is the process of digging the earth to provide a place for the foundation of the building.



2. Leveling and Grading are processes that change land elevation and slope by filling in low spots and shaving off high spots.

3. Stabilizing the Soil is the process of compacting the soil on which the structure will rest.



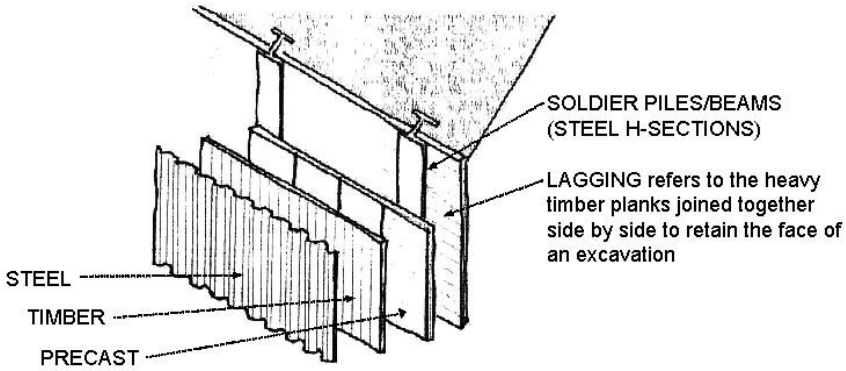
4. The protection of adjoining structure is a law that provides that any person making an excavation is responsible for resulting damage to adjoining property.

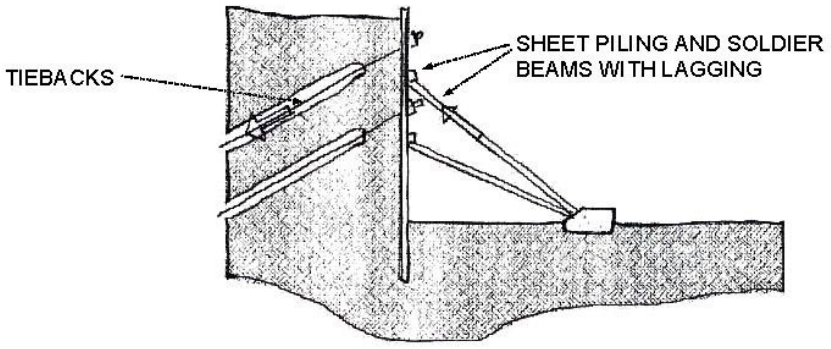
5. Shoring is a process of transferring a portion of the load of the wall to temporary footings and done when the excavation does not go much below the adjoining footings and when the material is fairly solid. It consists of either 1) **sheet piles,** which are timber, steel, or pre-cast planks driven side by side to retain earth and prevent water from seeping into the exaction and 2) **wales** or continuous horizontal beams which tie the sheet piles in place or 3) **soldier piles,** which are steel H-sections driven vertically into the ground to support 4) **lagging** or heavy timber planks joined together horizontally to retain the face of an excavation.

Crossbracing or **rakers** are diagonals which support the wales and soldier piles bearing on **heel blocks** or footings. **Tiebacks** secured to rock or soil anchors are resorted to when crossbracing or rakers would interfere with the excavation procedure.

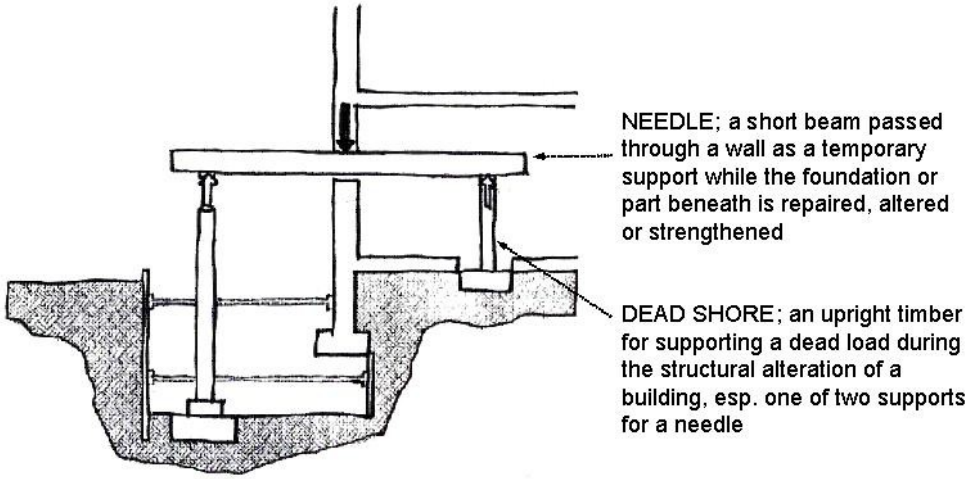




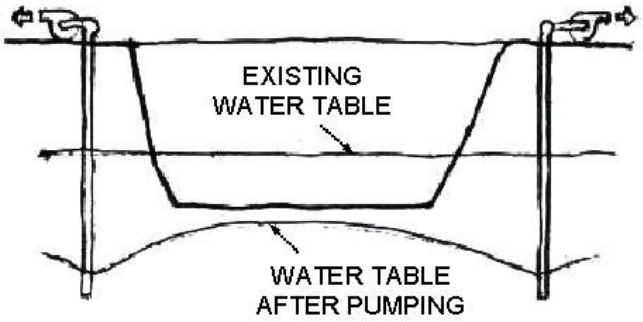




6. Needling and underpinning is a process where needles or girders are used in cases where part or all of the weight of the wall has to be carried, as when the old footing is removed and the wall underpinned or carried down to a new footing at a greater depth.



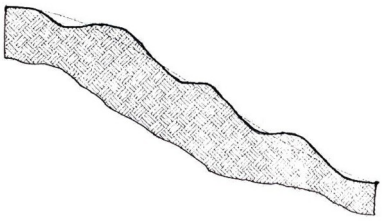
7. Dewatering refers to the process of lowering a water table or preventing an excavation from filling with ground water. It is accomplished by driving perforated tubes called well-points into the ground to collect water from the surrounding area so it can be pumped away.



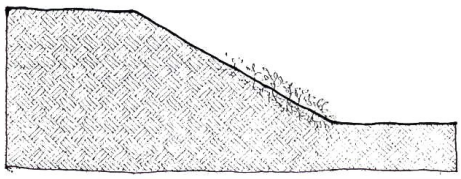
1.8. SLOPE PROTECTION and RETAINING STRUCTURES

Ground slopes over 25% are subject to erosion and may be difficult to build on. In order to protect the soil from eroding, the following slope protection systems may be done:

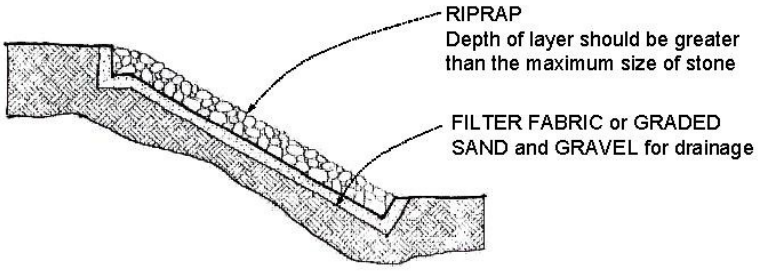
1. The need for stabilizing a sloping ground can be reduced by diverting the runoff at the top of the slope or by creating a series of terraces to reduce the velocity of the runoff.

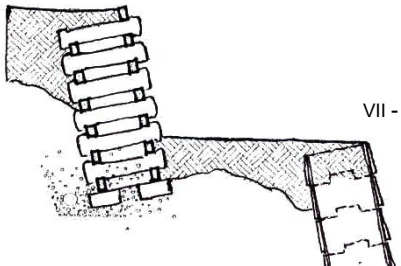


2. Natural means of stabilization include **soil binders** --- plant materials that inhibit or prevent erosion by providing a ground cover and forming a dense network of roots that bind the soil.



3. **Riprap** is a layer of irregularly broken and random-sized stones placed on the slope of an embankment.

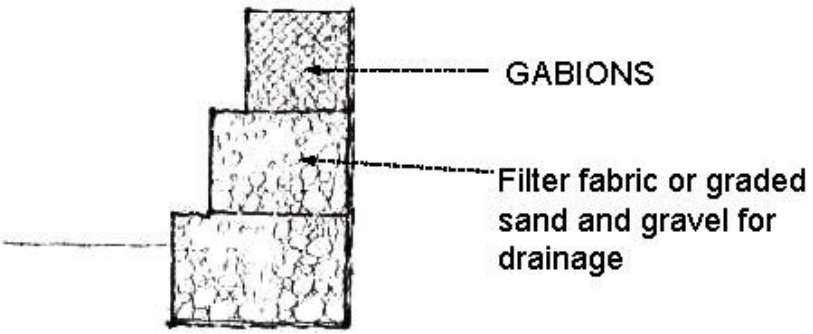


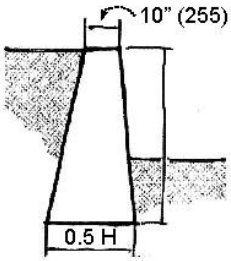


4. **Cribbing** is a cellular framework of squared steel, concrete, or timber members, assembled in layers at right angles, and filled with earth or stones.

5. A **Bin Wall** is a type of gravity retaining wall formed by stacking modular, interlocking pre-cast concrete units and filling the voids with crushed stone or gravel.

6. **Gabions** are galvanized or PVC coated wire baskets filled with stones and stacked to form an abutment or retaining structure.

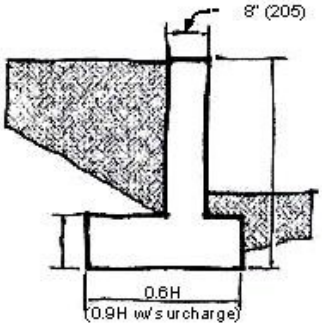


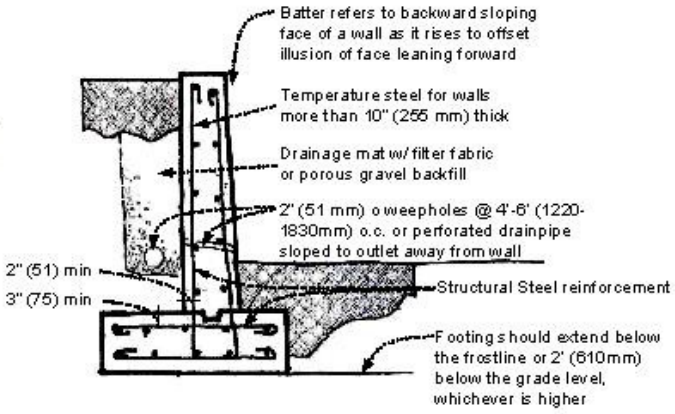


7. **Retaining Structures.** When a desired change in ground elevation exceeds the angle of repose of the soil, a retaining wall becomes necessary to hold the mass of earth on the uphill side of the grade change. The types of RC retaining walls are as follows:

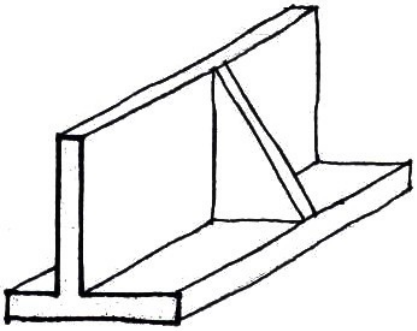
   a) **Gravity Retaining Wall –** resists overturning and sliding by the sheer weight and volume of its mass.

   b) **T-type Cantilevered Retaining Wall –** limited to a height of 20' (6 M); beyond this height a counterfort wall is employed.





   c) **Counterfort Retaining Wall –** utilizes triangular-shaped cross walls to stiffen the vertical slab and add weight to the base. The counterforts are spaced at equal intervals equal to one-half the wall height.



d) **L-type Cantilevered Retaining Wall –** used when the wall abuts a property line or other obstruction.

